

GitHub Repository Search System

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A search system for indexing and searching GitHub repositories with relevance-based ranking. The system automatically loads, indexes, and provides an interactive interface for searching a repository database.

Motivation

Modern developers rely on open-source repositories for learning, inspiration, and integration. However, finding the right project among millions on GitHub is challenging due to limited search and ranking capabilities. This project, GITGLE, addresses that gap by providing a distributed search engine that crawls, indexes, and ranks GitHub repositories based on README content and metadata. It enables fast, relevance-ranked search results, helping users discover high-quality repositories efficiently. The system supports scalable indexing, advanced ranking (TF-IDF, metadata), and interactive querying—making it a valuable tool for developers, researchers, and educators seeking the best open-source resources.

The system operates in three main phases, each distributed across nodes for scalability and fault tolerance:

1. Crawl Phase

- Loads a list of GitHub repositories from a seed file.
- Distributes repository URLs to nodes, which fetch metadata and README files from the GitHub API.
- Deduplicates and stores raw documents and metadata in a distributed key-value store.
- Tracks crawl progress and handles retries for failed fetches.

2. Index Phase

- Loads all crawled documents from storage.
- Runs multiple MapReduce jobs in parallel
- Extracts and normalizes terms (tokenization, stemming, n-grams).
- Calculates term frequency (TF), document frequency (DF), and inverse document frequency (IDF).
- Builds an inverted index mapping terms to document IDs, and stores all statistics in the distributed store

3. Search Phase

- User submits a query via the CLI.
- The query is parsed, normalized, and processed (tokenization, stemming).
- The system looks up matching documents in the inverted index.
- Calculates relevance scores using TF-IDF (75%) and metadata boosts (25%: stars, language, topics).
- Ranks results and formats them for display to the user.

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